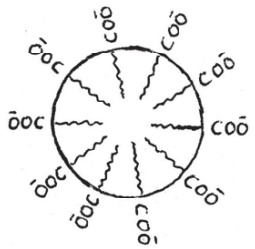
CHAPTER 17

CHEMISTRY IN DAILY LIFE



We use a number of chemical compounds - for example, water, salt, sugar, soap etc. - in our day to day lives. Would life be possible without water? Can you imagine how food would taste if we did not add sugar and salt to it? Not just salt but many other inorganic chemicals such as baking soda, baking powder etc. are used in food preparation and also in the baking industry. Bleaching powder is used to disinfect water and make it free of germs. Cement, glass, Plaster of Paris are used in construction of buildings. We will study many such important chemical compounds and mixtures in this chapter.

17.1 Water

Three-fourth of the earth's surface is covered with water and it is found in oceans, rivers, lakes etc. Water is also found up to a certain depth beneath the earth's surface. For all living beings (plants and animals), water is as essential as oxygen for survival. On earth, water is present in three forms. These forms are solid (ice), liquid (water) and gas (water-vapour).

Usually the density of a substance in the solid state is more than its density in the liquid state. At 4°C, the density of water in the solid state (ice) is 0.9 g/mL and its density in the liquid state is 1 g/mL. As you can see, at 4°C the density of ice is less than the density of water and therefore ice floats on water. This is a boon for water-life. In colder regions of earth, the surfaces of rivers and lakes freezes to ice. Since ice is a thermal (heat) insulator therefore, water below the ice-surface cools more slowly and does not freeze easily.

Water vapour present in air is known as humidity. Humidity levels can change from season to season. Water is capable of dissolving many-many substances and therefore it is also known as universal solvent.

17.1.1 Soft and Hard Water

Water is called hard water or soft water depending on the amount of salts present in it and also on how much lather it produces with soap.

Soft water: It quickly forms a lot of lather with soap. Rain water and distilled water are two examples of soft water.

Hard Water: It takes a long time to form lather with soap and very little lather is seen. Instead, hard water forms a precipitate, known as soap scum, with soap.

17.1.2 Causes of hardness of water

The presence of dissolved salts of magnesium or calcium causes hardness of water. The salts can be hydrogencarbonates or suphates or carbonates. If the dissolved salt is of sodium the water will be soft. Soap is a sodium salt of long chain fatty acids, for example, sodium stearate ($C_{17}H_{35}COONa$). It reacts with calcium and magnesium salts to give a precipitate. The reaction can be written as follows:

$$2C_{17}H_{35}COONa + M^{2+} \longrightarrow (C_{17}H_{35}COO)_2 M\downarrow + 2Na^{+}(aq)$$

Precipitate

Here, M^{2+} is being used for Ca^{2+} or Mg^{2+}.

Water will form lather with soap only after all the magnesium and calcium salts are precipitated. A lot of soap is wasted during the precipitation reaction. Additionally, the residual soap scum clings to clothes giving them a dull gray appearance.

17.1.3 Types of hardness

1. **Temporary hardness:** This is caused if dissolved calcium or magnesium hydrogencarbonates are present in water. When we boil water, the soluble salt (calcium hydrogencarbonate or magnesium hydrogencarbonate) gets precipitated as insoluble carbonate. You may have seen a white ring or white film deposited in utensils which are used for heating water; this is the insoluble carbonate.

$$Ca(HCO_3)_2(aq) \xrightarrow{\text{Boiling}} CaCO_3\downarrow + H_2O + CO_2\uparrow$$

Suppose the hardness of water is due to magnesium hydrogencarbonate. In this case, what will happen when we boil water? Can you write the chemical equation for this reaction?

Boiling is one method of getting rid of temporary hardness. Another way is by adding a fixed quantity of limewater to it.

$$Ca(HCO_3)_2(aq) + Ca(OH)_2 \longrightarrow 2CaCO_3\downarrow + 2H_2O(l)$$

2. **Permanent hardness:** This is caused by the presence of dissolved salts of calcium or magnesium chlorides and/or calcium and magnesium sulphates in water. Permanent hardness cannot be removed by boiling or treatment with limewater.

Modern techniques for softening hard water



Generally, two techniques are commonly used for softening of hard water that has excessive salts. One of the techniques is reverse osmosis (RO) method. You know that when two solutions are separated by a semi-permeable membrane then the solvent molecules of the dilute solution flow towards the concentrated solution through this membrane. This process is known as osmosis. But in reverse osmosis, high pressure is used so that water molecules from the concentrated solution (in this case, hard water) flow through the semi-permeable membrane and we get pure water on the other side. Thus, a semi-permeable membrane is used to separate salt molecules from hard water. However, a lot of water is wasted during reverse osmosis purification.

In the second technique, water is softened using ion-exchange resin (where the resin is porous). A resin with sodium ions is used to filter hard water. When the hard water passes through the resin, Ca^{2+} and Mg^{2+} ions present in the water get exchanged with Na^{+} ions. The water thus obtained is soft as it is free from Ca^{2+} and Mg^{2+} ions. The Na^{+} ions in the resin decrease with use and the resin has to be replenished from time to time.

17.1.4 Potable water

Drinking water should be clean, colourless, odourless and free from germs. It should also have a sufficient amount of dissolved salts. To make the available water potable, we can add alum, bleaching powder, potassium permanganate, chlorine etc. as needed to water purification devices. We can use ultraviolet rays to purify water.

Distilled water, which is obtained by distillation, is used in medical research, car batteries and in experimental work.

Try to use solar energy to prepare distilled water for your school's laboratory.

17.1.5 Uses of water

1. Water is used as a solvent.
2. Water is essential for crops and in agriculture.
3. Water is used in a number of industries such as textile industry, paper industry, mining etc.
4. Water is used to wash clothes and for cleaning purposes.
5. Since water has high specific heat, it is used in boiler engines, coolers etc.

17.1.6 Water pollution

When water is used (for cleaning or in industries), some undesirable substances get mixed in it. Further, there can be decrease in amount of oxygen dissolved in water due to many reasons and this can affect the properties of water. This water is now polluted and is no longer suitable for use.

You must be familiar with many causes of water pollution; list some factors that cause water pollution in your area. It is our responsibility to prevent water pollution and control it. Some ways of preventing and limiting water pollution are:

1. Making it compulsory for all industries to install water purification devices to treat waste water produced by them.
2. Household waste water and sewage water should be treated before it enters rivers and lakes.
3. The use of harmful pesticides should be limited and their excessive use should be banned.
4. Many human activities, such as bathing; washing clothes; immersing dead bodies, ashes of cremated persons, idols, flowers etc. in rivers; also pollutes water. Such activities need to be monitored and controlled.

17.2 Common Salt

Sea water has 30-35 g of dissolved salts per liter of water. The primary salt present is sodium chloride which is also known as common salt.

17.2.1 Salt production

Sea water is collected in shallow ditches near the seashore and left for evaporation. Once the water evaporates it leaves behind raw salt, which contains many other salts and a small quantity of sand along with common salt (figure-1). This raw salt is purified and crystallized for use. Obtaining salt from sea water is difficult and the task has to be carried out very carefully. During the purification process of raw salt, some desired and necessary substances are also added to it for example, iodine. Iodine is added in the form of potassium iodate. Can you tell why iodine is added to common salt?



Figure-1: Production of salt from sea water

A few regions of the world have salt mines that give us rock salt. It is believed that some parts of seas or oceans must have dried up long ago and resulted in these salt mines.

Common salt is an essential component of our diet but can we use sea water to cook food?

Salt plays an important role in several biological processes. Therefore, our food should have the right quantity of salt in it. High amounts of salt in our body can cause high blood pressure and low amounts may lead to low blood pressure.

Do you know that many packaged foods such as chips and *namkeen* have very high amounts of salts and are therefore harmful?

17.2.2 Uses of salt

1. Common salt is used as a preservative that prevents food from spoiling by stopping the growth of bacteria.
2. Common salt is an important raw material in production of many substances, such as sodium hydroxide, baking soda, washing soda, bleaching powder etc.
3. Oral rehydration solution (ORS), which is a solution of sugar and salt in water, is the primary treatment for when our bodies get dehydrated.

Questions

1. We have two water samples, 'A' and 'B'. Sodium hydrogencarbonate was found in sample 'A' and magnesium sulphate was found in sample 'B'. Which sample of water is hard and why?
2. Give two natural sources of common salt.

17.3 Baking Soda

Baking soda is often used in the kitchen to obtain spongy cakes and breads. Sometimes, it is also used to cook food faster. Its chemical name is sodium hydrogencarbonate and it is also known as sodium bicarbonate.

Baking soda is produced by Solvay ammonia process which uses salt as one of the raw materials.

$$NaCl + H_2O + CO_2 + NH_3 \longrightarrow \underset{\textit{Ammonium chloride}}{NH_4Cl} + \underset{\textit{sodium hydrogencarbonate}}{NaHCO_3}$$

In a previous chapter, you checked the pH value of sodium hydrogencarbonate. Can you tell why it is used to neutralize acidic solutions?

When sodium hydrogencarbonate is heated it decomposes to form sodium carbonate with production of carbon dioxide gas.

$$2NaHCO_3 \xrightarrow{\text{Heat}} Na_2CO_3 + H_2O + CO_2$$

17.3.1 Uses of baking soda

1. In making baking powder – Baking powder is a mixture of baking soda and tartaric acid. When baking powder is heated or mixed with water, the following reaction takes place–

$$\underset{\textit{(from acid)}}{NaHCO_3 + H^+} \longrightarrow CO_2\uparrow + H_2O + \text{Sodium salt of acid}$$

   Carbon dioxide produced during the reaction causes bread or cakes to rise, making them soft and spongy.

2. Sodium hydrogencarbonate is used as an antacid to reduce acidity of stomachs. Being alkaline, it neutralizes excess acid in the stomach.

3. Baking soda is also used in soda-acid fire extinguishers.

17.4 Washing Soda

The chemical formula of washing soda is $Na_2CO_3.10H_2O$. We have seen that sodium carbonate can be obtained by heating baking soda and recrystallization of sodium carbonate gives washing soda. Washing soda is also a basic salt.

$$Na_2CO_3(s) + 10H_2O(l) \longrightarrow Na_2CO_3.10H_2O(s)$$

Can you tell: What does $10H_2O$ signify? Does it make Na_2CO_3 wet?

10 molecules of water are attached to each molecule of crystalline sodium carbonate. This does not mean that it is in the form of a solution. Washing soda is a white-coloured crystalline solid with attached water molecules. Water of crystallization is the fixed number of water molecules presentin one formula unit of a salt. When washing soda is left in open air, it loses nine water molecules and gets converted into a white powder of sodium carbonate monohydrate. This property is known as efflorescence.

$$Na_2CO_3.10H_2O(s) \xrightarrow{\text{left in open}} Na_2CO_3.H_2O(s) + 9H_2O$$

17.4.1 Uses of washing soda

1. Sodium carbonate is used as a cleaning agent for domestic purposes.

2. It is used in glass, soap, textile and paper industries.
3. It is used for removing permanent hardness of water.

Questions

1. Write the chemical formula and two important uses of baking soda.
2. What is efflorescence?
3. Baking powder is a mixture of which two compounds?

17.5 Plaster of Paris

Chemically, Plaster of Paris is calcium sulphate hemihydrate. Its chemical formula is $CaSO_4.½H_2O$ or $(CaSO_4)_2.H_2O$

It is formed by heating gypsum at 100°C.

$$\underset{\text{gypsum}}{CaSO_4.2H_2O} \xrightarrow{100°C} \underset{\text{Plaster of Paris}}{CaSO_4.½H_2O} + 1½H_2O$$

Plaster of Paris is a white powder and on mixing with water, it changes once again to gypsum giving a hard solid mass.

$$CaSO_4.½\ H_2O + 1½\ H_2O \longrightarrow CaSO_4.2H_2O$$

17.5.1 Uses of Plaster of Paris

1. It is used by doctors for supporting fractured bones in the right position.
2. It is used for making toys and idols.
3. Used to make casts and models.
4. Used in making chalk.

When idols made of Plaster of Paris are immersed in lakes, ponds and rivers, they pollute the water source. In your classroom, form groups and discuss how these idols cause water pollution. These days, to counter this problem of water pollution, people are encouraged to buy idols made of clay instead.

17.6 Bleaching powder

Chemically, bleaching powder is known as calcium oxychloride ($CaOCl_2$). It is also referred to as chloride of lime. Bleaching powder is produced by the action of chlorine gas on dry slaked lime.

$$\underset{\textit{slaked lime}}{Ca(OH)_2(s)} + Cl_2(g) \longrightarrow \underset{\textit{bleaching powder}}{CaOCl_2(s)} + H_2O(g)$$

It is a white powder having a yellowish tinge with a strong smell of chlorine. When left in the open, it reacts with carbon dioxide to release chlorine gas.

$$CaOCl_2(s) + CO_2(g) \longrightarrow CaCO_3(s) + Cl_2(g)$$

17.6.1 Uses of bleaching powder

1. For bleaching cotton in the textile industry. This bleaching action is due to the chlorine gas released by bleaching powder.
2. For bleaching wood pulp in paper factories.
3. For disinfecting drinking water to make it free of germs.

Questions

1. Which property of Plaster of Paris is the reason why doctors use it in setting fractures?
2. Name the substance that reacts with chlorine to give bleaching powder.

17.7 Cement

Cement is essential in construction of buildings. The meaning of the word cement is something that binds or glues together other materials. Cement was first made by Joseph Aspidin. He called it Portland cement because it was as hard as the limestones found in Portland.

Cement is not a chemical compound. It is a mixture of chemical compounds such as silicates and aluminates of calcium. Therefore, it does not have a fixed chemical composition or formula. The properties of cement are also not fixed, for example, it does not have a fixed melting point. The percentage composition of cement is shown in table-1:

Table-1 : Percentage composition of different components in cement

Substance	Formula	Percentage composition
Calcium oxide	CaO	61-65%
Silica	SiO_2	20-25%
Alumina	Al_2O_3	5-10%
Magnesia	MgO	2-3%
Iron oxide	Fe_2O_3	1-2 %

When the fine powder of this mixture comes in contact with water, it sets into a hard substance.

17.7.1 Cement manufacture

The raw materials needed in manufacture of cement are limestone ($CaCO_3$), clay and gypsum. Clay contains large quantities of oxides of iron, aluminium and silicon. The steps in cement production are as follows:

(i) Three parts limestone and one part clay are pulverized to fine powder separately. The powders are mixed and heated in a kiln where temperature is maintained between 1100-1800°C (figure-2).

(ii) At 1100°C, limestone undergoes thermal decomposition and forms calcium oxide.

$$CaCO_3 \xrightarrow{1100°C} CaO + CO_2$$

(iii) Calcium oxide reacts with the other oxides present in clay.

$$2CaO + SiO_2 \xrightarrow{1400\text{-}1500°C} 2CaO.SiO_2$$

Dicalcium silicate

$$3CaO + SiO_2 \xrightarrow{1400\text{-}1500°C} 3CaO.SiO_2$$

Tricalcium silicate

$$3CaO + Al_2O_3 \xrightarrow{1400\text{-}1500°C} 3CaO.Al_2O_3$$

Tricalcium aluminate

$$4CaO + Al_2O_3 + Fe_2O_3 \xrightarrow{\text{above } 1500°C} 4CaO.Al_2O_3.Fe_2O_3$$

Tetracalcium alumino ferrite

(iv) The aluminate and silicate mixture obtained from the furnace is in the form of small, hard balls or pebbles and is called clinker.

(v) Clinker is cooled and 2-3% gypsum ($CaSO_4.2H_2O$) is added to it. This mixture is pulverized into a fine powder.

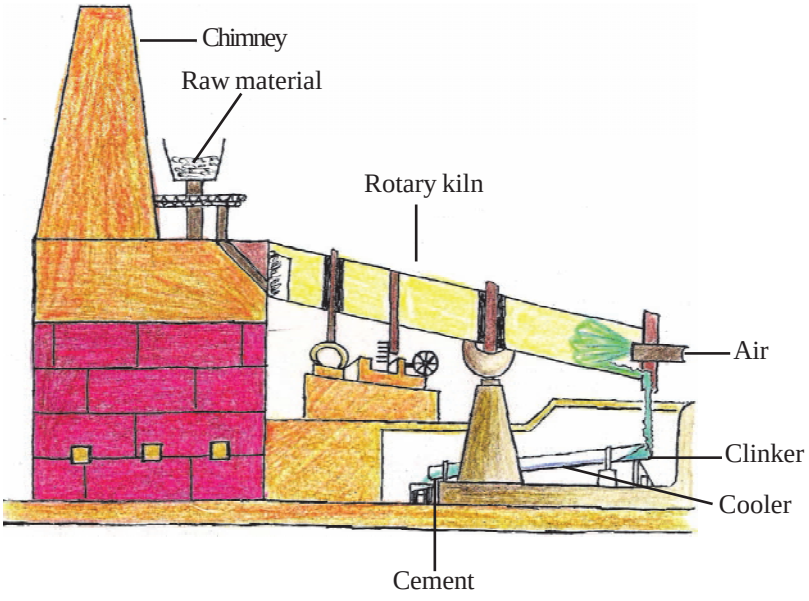
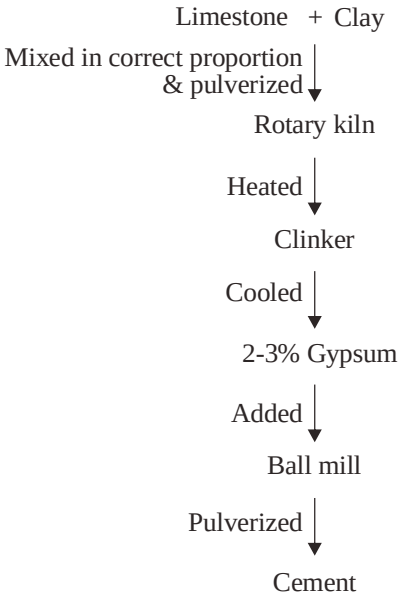


Figure-2: Manufacture of Cement



Flow chart-1: Method for cement preparation

17.7.2 Setting of cement

When cement is mixed with sand and water and left for some time, it sets and becomes as hard as a rock. Gypsum is added to cement to slow down the setting process.

The demand for cement is increasing day by day and to reduce the cost of cement manufacture, these days industrial waste

materials such as fly ash are used to produce cement. In some countries, rice bran ash is also used to make cement because it has high quantities of silica.

17.7.3 Cement industries in Chhattisgarh

Cement factories in Chhattisgarh are located in the following districts- Raipur, Bilaspur, and Ballaudabazaar.

17.8 Glass

Glass is a mixture of silica and metal silicates. It is non-crystalline, hard, brittle, and transparent and appears solid. In reality, glass is a super-cooled liquid which looks solid but is capable of flowing.

The general formula of glass is: $xR_2O.yMO.6SiO_2$where

R is a monovalent alkali metal such as Na, K etc. and M is a divalent metal such as Ba, Ca, Pb, Zn etc. x and y indicate number of molecules.

17.8.1 Manufacture of glass

The materials used in manufacture of glass are:

1. Silica: in the form of sand.
2. Alkali metal: in the form of sodium carbonate or potassium carbonate or mixture of salt cake and carbon (Na_2SO_4 + C).
3. Divalent metal: such as calcium in the form of limestone or lead in the form of litharge (PbO) or red lead oxide (Pb_3O_4).
4. Oxidizing agents or materials to remove colour (bleaching agents): manganese dioxide (MnO_2), potassium nitrate (KNO_3) or sodium nitrate ($NaNO_3$).
5. Cullets: pieces of broken glass
6. Colouring materials: Various compounds used to impart different colours to glass (table-2).

Table-2: Colouring materials

Colour of glass	Colouring material
Green	chromium oxide
Yellow	cadmium sulphide
Red	copper oxide
Purple	manganese oxide
Blue	cobalt oxide

To manufacture common glass or soda glass, the necessary raw materials - sodium carbonate, calcium carbonate and sand - are mixed in the right proportion and ground finely. Then cullet, which acts as flux, is added to this mixture. The mixture thus obtained is called batch. Batch is heated upto 1400°C (figure-3) in a tank furnace. The following reactions take place in the furnace:

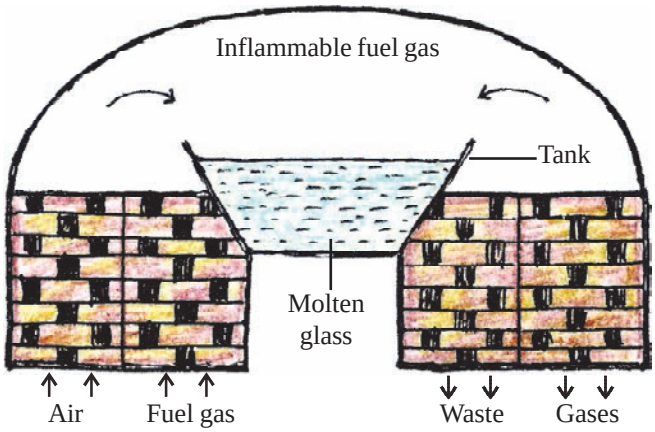
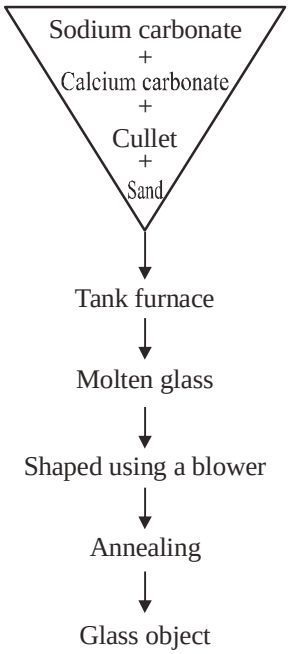


Figure-3: Manufacture of glass in tank furnace



Flow chart-2: Steps for making glass

$$Na_2CO_3 + SiO_2 \longrightarrow Na_2SiO_3 + CO_2\uparrow$$

$$CaCO_3 + SiO_2 \longrightarrow CaSiO_3 + CO_2\uparrow$$

$$Na_2SiO_3 + CaSiO_3 + 4SiO_2 \longrightarrow \underset{\textit{glass}}{Na_2O.CaO.6SiO_2}$$

Experienced craftsmen can shape this molten glass into desired shapes and objects using a blower. These days this is done in big factories where machines use moulds and the process takes less time and costs less. Glass can crack and break if cooled suddenly during production. To prevent this, glass is cooled very slowly and the process is called annealing.

Questions

1. Why is gypsum added to cement?
2. Why are cullets used in glass manufacture?
3. The melting point of glass is not fixed, why?

17.9 Soaps and detergents

Activity-1

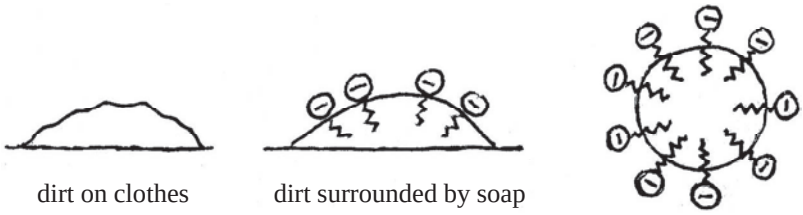
- Take two test tubes and label them as 'A' and 'B' respectively.
- Take 10 mL of distilled or soft water in each test tube.
- To each test tube, add 4 drops of cooking oil.
- Add a few drops of soap solution to test tube 'B'.
- Shake both the test tubes vigorously for the same time duration.
- Look for separate water and oil layers in both the test tubes immediately after you stop shaking them. Do you see anything?
- Leave the test tubes undisturbed. Did the layer of oil separate out from the water? In which test tube did it separate out first?

This activity shows the role of soap in cleaning.Usually, dirt is oily in nature and we know that oil does not dissolve in water. The molecules of soap are sodium or potassium salts of long chain carboxylic acids and are denoted by RCOOM. When soap is dissolved in water, it ionizes into $RCOO^-$ and metal, M^+ ions. The $RCOO^-$ ion has two parts. One is R which is a long hydrocarbon chain that forms the tail of the carboxylate ion and gets attached to the oil molecules. The second part is COO^- which forms the head and is soluble in water (figure-4a).

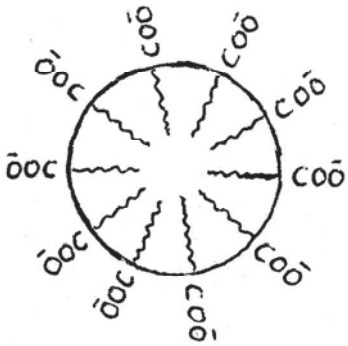
When clothes are soaked in soap solution, then the negatively charged part of soap surrounds the oil (dirt). A sphere is formed around the oil drop in such a way that the end with the hydrocarbon chain (tail) is inside the sphere and the COO^- end is on the outer part of the sphere. Thus, the soap molecules form micelles (figure-4b). The micelles get distributed in water forming a colloid which is removed when we rub the cloth and rinse with water (figure-4c).



4 (a): parts of carboxylate ion



4 (b): COO^- ion arrangement



4 (b): action of soap in removal of dirt

Figure-4 (a), (b) and (c): Detergent micelle

Activity-2

- Take about 10 mL of distilled water and 10 mL of hard water in two separate test tubes.
- Add a few drops of soap solution to both.
- Shake the test tubes vigorously for an equal period of time and observe the amount of foam formed.
- In which test tube do you get more foam?
- In which test tube do you observe a white curdy precipitate?
- What can you conclude from this activity?

Note for the teacher: *If hard water is not available in your locality, prepare some hard water by dissolving magnesium hydrogencarbonate/chloride/sulphate in water.*

In this chapter, we have already discussed hardness of water. To clean objects using hard water, we can use another class of compounds called detergents. Detergents are salts of sulphonic acids having along carbon chain. The charged ends of these compounds do not form insoluble precipitates with the calcium and magnesium ions in present hard water. Thus, they remain effective in hard water. Detergents are usually used to make shampoos and other products for cleaning fabrics.

Activity-3

- In two separate test tubes, take about 10 mL hard water each.
- Add five drops of soap solution to one and five drops of detergent solution to the other.
- Shake both test tubes for the same period.
- Do both test tubes have the same amount of foam?
- In which test tube is a curdy solid formed?
- What can you conclude from this activity?

In this activity, we compared the cleansing action of soap and detergent in hard water. Some detergents are non-biodegradable and thus increase water pollution; they should be used in moderation.

Questions

1. Can you use detergents to determine if a given sample of water is hard or not?
2. When washing clothes, after applying soap we either scrub with a brush or beat clothes over a stone or beat them with a stick. Why do we need to scrub clothes to clean them?

Keywords

Efflorescence, bleaching powder, clinker, micelle, gypsum, solid to touch, supercooled liquid, detergent, cullets, temporary and permanent hardness, annealing



What we have learnt

- Water is capable of dissolving more substances than any other solvent and therefore it is also known as universal solvent.
- The presence of dissolved salts (hydrogencarbonates or suphates or carbonates) of magnesium or calcium causes hardness of water.
- Common salt is iodized by adding potassium iodate to it.
- Baking soda is produced by Solvay ammonia process using sodium chloride as one of the raw materials.

- When washing soda crystals are left in open air, each crystal loses nine water molecules and gets converted into a white powder of sodium carbonate monohydrate;this property is known as efflorescence.
- Plaster of Paris on mixing with water hardens into gypsum.
- Bleaching powder is used in bleaching. This property is due to the chlorine gas produced by it.
- Cement is mostly a mixture of silicates and aluminates of calcium which hardens when it comes in contact with water.
- Glass is a mixture of metal silicates and therefore does not have a fixed composition or chemical formula.
- Soaps are sodium or potassium salts of long chain carboxylic acids.
- Detergents are usually salts of long carbon chain sulphonic acids.

Exercises

1. Choose the correct option:

   (i) The chemical formula of baking soda is:

   (a) $NaHSO_4$ (b) Na_2CO_3

   (c) $NaHCO_3$ (d) $Na_2CO_3.10H_2O$

   (ii) The chemical substance that is used for disinfecting drinking water to make it free of germs is:

   (a) $CaCl_2$ (b) $CaOCl_2$

   (c) $FeCl_3$ (d) $MgCl_2$

   (iii) Glass is a

   (a) Liquid (b) Solid

   (c) Transparent carbon polymer (d) Super-cooled liquid

   (iv) Plaster of Paris hardens by

   (a) Losing $CaCl_2$ (b) Releasing CO_2

   (c) Absorbing water (d) Releasing water

2. Match words in column A with the correct options in column B

Column A	**Column B**
Bleaching powder	super-cooled liquid
Baking soda	clinker
Washing soda	preservative
Sodium chloride	Bleaching agent
	Glass manufacture
	antacid

3. What is hard water? What is soft water?
4. What are the factors that cause hardness of water? How many types of hardness are there?
5. How do animals and plants, living in lakes and ponds in snowy regions, survive during winter?
6. What is baking powder? How does it make cakes soft and spongy?
7. What is the process of making Plaster of Paris? Why is it stored in air-tight containers?
8. What is the chemical name for washing soda? Name the three main raw materials used to produce washing soda by the Solvay ammonia process.
9. A compound of calcium, which is a white powder having a yellow tinge, is used in the textile industry and as a disinfectant.
   (i) Name the compound.
   (ii) Which gas is released when the compound is left out in open air? Give the balanced equation for this reaction.
10. A compound of sodium, 'X', is a white powder and an important component of baking powder. When 'X' is heated, a gas 'Y' is produced that turns lime water milky.
    (i) Write the chemical equation for the reaction which takes place on heating.
    (ii) Why is 'X' used as an antacid?
11. Describe the steps in the process of glass manufacture under each of the following headings:
    (i) Necessary materials
    (ii) Substances that give or impart colour to glass
    (iii) The chemical reaction taking place in tank furnace
12. Write the different steps involved in cement production.
13. How are soap-micelles formed? Describe.